

Kyle Christian Chong

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EDUCATION

University of Texas at Austin | GPA: 3.47

Graduation: Fall 2027

B.S. Mechanical Engineering

EXPERIENCE

UT Austin (NDML) - Undergraduate Semiconductor Equipment Researcher

January 2026 - Present

- Designed a flexure-based photomask mount for **contact lithography** to minimize optical mask tip/tilt, achieving 155 μrad predicted pre-contact tilt under gravity; validated deflection and stress in SolidWorks FEA against calculated 209 μrad requirement.
- Derived an error budget allocating tilt contributors across the assembly, yielding a $\pm 0.015''$ manufacturing requirement; specified the assembly procedure and built components via CNC milling and sheet-metal shearing.
- Cleanroom-trained in safety, contamination control, and standard gowning/handling procedures.

Imagine Devices - Associate R&D Engineer

March 2026 - Present

- Built and documented the analog pressure-sensor system for the company's flagship medical device, quantifying the error budget of ± 0.06 inH₂O now used by the team to quantify pressure readings.
- Designed and built a benchtop neonatal breathing simulator to document pressure-sensing capability in a medical setting built to support upcoming regulatory design verification.
- Designed the experimental setup for thermistor validation by characterizing thermal noise caused by an adjacent feeding lumen.

ENGINEERING PROJECTS

Articulated Mechatronic Wings

May 2026 - Present

- Synthesized 7-bar pantograph linkage that unfolds a 4.5-ft wing from a single actuator, using a custom Python solver to reduce 135 candidate solutions to the final mechanism.
- Sized actuator by deriving a quasi-static holding torque of 3.9 N·m per wing (7.8 N·m total) via the Principle of Virtual Work, using SolidWorks mass properties.
- Selected materials via a weighted decision matrix, specifying Carbon Fiber (CFRP) for high-shear central links and balsa (4× lighter) for outer links to maximize strength-to-weight.

Snap-Detecting Watch: Hardware & Enclosure Design

May 2025 - June 2026

- Architected a 4-layer wearable PCB (KiCad), isolating 2.4GHz RF routing from continuous-listening analog microphone traces to ensure high-fidelity acoustic signal integrity.
- Performed ANSYS structural analysis to quantify PCB stiffener effects, driving four physical design-test iterations that implemented analog filtering to eliminate enclosure-induced resonance and acoustic signal noise.
- Designed thin-walled enclosure revisions in SolidWorks, executing rigorous material-specific tolerance analysis (PLA/Resin) to ensure robust packaging for the battery, sensors, and PCB across multiple builds.

Snap-Detecting Watch: Real-Time Machine Learning Classifier

May 2025 - June 2026

- Engineered an embedded DSP pipeline to extract time-domain acoustic features (ZCR, Crest) and PCA-reduced spectrograms from 2048-sample audio windows to prepare data for real-time classification.
- Evaluated classification models via exhaustive feature selection and hyperparameter grid searches, benchmarking linear against non-linear decision boundaries using a dataset of 300+ custom gesture samples.
- Trained a Random Forest model to F1 = 92.3%, prioritizing high trigger recall and minimal false-positives to ensure reliable gesture detection natively on an nRF52833 wearable device.

TECHNICAL SKILLS/CERTIFICATIONS

- CAD/FEA/Fab: SolidWorks, ANSYS, GD&T, 3D printing, hand reflow/drag soldering
- Test / Prototyping: LabVIEW, NI DAQ/data acquisition, oscilloscopes, digital multimeters, thermocouples, calipers
- Data Analysis (Cleaning + Visualization): Excel, Python (NumPy, Pandas), MATLAB
- Embedded: KiCad, STM32CubeIDE (C), analog sensors, digital signal processing
- Six Sigma/Lean Yellow Belt Coursera Certification